



Maritime &
Coastguard
Agency

Guidance

Annex 1 - Competence Standards for General Towage Endorsement

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A1.1 Summary

This standard is about:

The competence required to conduct general towage operations (towing and pushing) at a management level on board vessels operating in categorised waters or in limited coastal areas.

This standard is for:

Masters of vessels engaged in general towage operations (towing and pushing) in categorised waters or in limited coastal areas.

A1.2 Outcomes of effective performance

Candidates for the issue of the General Towage Endorsement will have demonstrated that they can do the following:

1. Ensure that there are clearly defined safe work procedures/standing orders for conducting safe general towage operations.
2. Inspect the vessel to be towed and establish correctly that it is safe to tow to the planned destination.
3. Inform the relevant authority promptly where you assess the vessel to be unfit for towing, stating the reasons clearly.
4. Where necessary, ensure that all statutory, regulatory and other requirements for the tow are identified and met, and comply with all navigational, reporting and warning requirements.
5. Where the vessel is subject to compulsory pilotage, ensure there is an adequate exchange between the Master of the towing vessel and Pilot and effective bridge team support for the Pilot.
6. Assess the circumstances affecting the requirement for a tow and determine correctly any risks involved.
7. Plan the tow, identifying correctly the actions required, taking into account the characteristics of the tow, and any relevant navigational restrictions, determining the best method for towing.
8. Establish and maintain clear and appropriate communications between your vessel and that to be towed.
9. Brief all concerned clearly and correctly on the actions to be taken, ensuring that everyone understands their role and the emphasis upon

safety.

10. Prepare correctly all necessary equipment for towing and rigging, checking for damage prior to taking a tow, making good and reporting any faults before use.
11. Develop and action appropriate contingency plans that ensure the safety of navigation, protection of the marine environment and safety of the vessel and persons on board.
12. React safely and appropriately to loss of critical systems e.g. loss of main power or steering unit whilst engaged in general towage operations.
13. Identify where the transfer of a crew member to the vessel to be towed would assist the towing operation, and where relevant, undertake such a transfer safely and correctly.
14. Rig, secure and deploy the towing equipment correctly, taking necessary actions to protect both vessels and to minimise the effect of chafe.
15. Ensure that the correct audible and visual signals indicate towing.
16. Handle the vessel safely under the prevailing conditions with due consideration to the engineering systems.
17. Operate remote controls of propulsion plant, engineering systems and services correctly and safely.
18. Monitor the tow and make any necessary adjustments correctly and safely.
19. Establish communications with those receiving the tow and other relevant authorities and ensure that preparations for receiving the tow on arrival are in place.
20. Disengage the tow correctly and safely.

A1.3 Knowledge and understanding

Underpinning Knowledge Syllabus for General Towage Endorsement
Candidates need to know and understand the following:

GT 1 Internal Communications

The use of internal communications systems and effective forms of communications including:

- 1.1 Pre-tow briefing/ toolbox talks
- 1.2 Use of hand signals
- 1.3 Use of hand held radios
- 1.4 Use of CCTV
- 1.5 On board alarms/announcements/signage

GT 2 External Communications

External communications with other vessels, pilots and coastal authorities including:

- 2.1 Tow set up briefing
- 2.2 Use generic terminology after confirming common understanding with pilots and shore authorities
- 2.3 Use of the IMO Standard Marine Communication Phrases (SMCP) as appropriate
- 2.3 Use of radios and commands/instructions
- 2.3 Use of other means of communications including AIS
- 2.4 Reports to navigation authorities/Harbour Master/VTs/Port Control/HM Coastguard

GT 3 Emergency Procedures

The actions to be taken in the event of:

- 3.1 Failure of towing lines
- 3.2 Failure of Gog arrangements/ shackles/ towing hook/ winch
- 3.3-5 Failure of critical systems including – engine/ steering/ propulsion/ electrical
- 3.6 Loss of communication with towed vessel/ pilot/ port control
- 3.7 Mechanical problem on towed vessel including engines/ steering/ other (e.g. release or loss of anchors)
- 3.8 Rope in propulsion system
- 3.9 Failure of watertight integrity of tug when towing

- 3.10 Collision
- 3.11 Grounding of tug and/or tow
- 3.12 Man overboard
- 3.13 Fire
- 3.14 Pollution
- 3.15 Use of emergency controls
- 3.16 Deployment of emergency tow line/ emergency anchor systems
- 3.17 Emergency release of tow
- 3.18 Deployment of crew in emergency situations
- 3.19 Constraints on ability to render assistance to others/ SOLAS statutory obligations
- 3.20 Assessment of risks/ limitations on ability of voluntary assistance for salvage

GT 4 Fitness for Purpose

Ensuring that:

- 4.1-2 Vessel has appropriate certification for tug and tow.
Vessel has appropriate lights and daylight shapes for towage available
- 4.3-4 Navigational equipment is in keeping with working areas
- 4.5 There is sufficient crew, suitably qualified and experienced for the intended work
- 4.6 Watertight integrity is maintained during towage operations
- 4.7 Vessel is of a size and type with power and bollard pull appropriate to carry out work intended and expected conditions.
- 4.7 All lifting and towing equipment is tested and/ or certified and of a size and type consistent with type of work and operational area.
- 4.8 You are aware of meaning and purpose of the term 'Warranty Survey'.

GT 5 Local knowledge and passage planning when towing and pushing

Construct a passage plan for a voyage and demonstrate to the master that you have taken account of the relevant local conditions, including:

5.1 Effect of local conditions on tows e.g., wind, tide, depth, localised tidal effects 'run off', interaction (squat, canal effect) effect of propeller wash, currents and berths

5.2 Local traffic conditions, pilotage and port movements

5.3 Passage planning - to note the effect of wheel over on tow position

5.4 Track of tow in narrow channels

5.5 Ports of refuge and safe havens

5.6 Use of additional tugs for critical points in the passage

5.7 Grounding between tides (ebbing up)

GT 6 Safe Towage Operation

How to plan a towage operation taking into account:

6.1 Identification of the principal risks and method of assessment

6.2 Type and characteristics of towing gear to be used

6.2 Determining the towing methods to be used

6.2 Double headed tow/ tandem tow/ integrated tug systems for pusher tugs/ non integrated pushing/ Mississippi arrangements

6.3-4 The inspection of the tow, assessment of suitable towage points and chafing areas

6.5 Close quarters operations – effect of interaction, wash, restricted waters

6.5 Characteristics of the tow

6.6 Rigging, correct deployment and safe handling of the towing gear

6.7&15 Connecting up and letting go, changing towage arrangements

6.8 Safe areas on deck

6.9 Sufficient lighting – deck lighting/directional searchlight to illuminate tow

6.10 Stability for towing operations

6.12 Relevant local byelaws

6.13 Appropriate towage arrangements for different phases of the tow

6.14 Berthing arrangements

On passage

6.16 Monitoring the tow, and taking safe and timely corrective action if required

6.16 Knowledge of towing points. Risk and prevention of girting - e.g. use of towing pins, gog ropes etc

6.17 Avoidance of large dynamic tensions ('snatching') in the towline

GT 7 Towing Equipment

How to use loose equipment safely taking into account:

7.1 Safe Working Loads - knowledge of difference between lifting regulations and towing industry standards

7.2 Types of ropes, wires and chains commonly used in towage

7.3 Characteristics, monitoring, care, testing and certification of all loose equipment, proof tests and manufacturers certificates

7.4 Towing Bridles, stretchers and chafe chains, gogs, gobs, bridles and V ropes

Types of shackles, uses, and securing

Monkey plates (face plates)

Stoppers

Towing pins, Shark jaws and other line restraining devices

Anti-chafe devices

How to use and maintain the following fixed equipment safely taking into account their limitations:

7.5-9 Bollards, bitts and samson posts

Winches and capstans, including controls, brakes and tension measuring devices

7.10-12 Towing hooktypes - release mechanisms, maintenance and testing procedures

7.13 Fairleads, rollers and towing brackets e.g. Smit bracket

7.14 Fendering – types and securing methods

GT 8 Propulsion systems and tug types

Have knowledge of operational characteristics of the following range of propulsion and steering systems:

8.1 Azimuth propellers - 360° steerable propellers which can deliver thrust in any direction (Z pellers)

8.2 Controllable pitch propeller(s)(CPP)

8.3 Fixed pitch propeller(s)(FPP)

8.4 Voith Schneider (VS) propulsion – vertical propeller blades

8.5 Steerable nozzles

8.6 Shrouded nozzles e.g. Kort type

GT 9 Differentiate the following tugs in terms of towing point with respect to propulsion system

Have knowledge of the classification of tug types in terms of the position of their propulsion systems with respect to their towing point:

Propulsion forward of midships with a towing point aft:

9.1 Tractor Tugs with Voith (VS)

9.2 Tractor tugs with Azimuth propellers

Propulsion aft and towing point near midships:

9.3 Conventional type – single or twin FPP or CPP

Intermediate tug types – dependent on method of operation:

9.4 Reverse tractor or pusher tugs

9.5 Combi tugs – modified older tugs with a 360° steerable thrusters in the bow.

9.6 Azimuth stern drive (ASD)- usually twin units

GT 10 Describe the main characteristics of the following vessel types and functions

Have a general knowledge of classification of tugs according to function:

- 10.1 Seagoing tugs
- 10.2 Escort tugs – passive or active
- 10.3 Harbour/ship assist tugs/ Carousel tugs
- 10.4 Anchor handling tugs
- 10.5 Craft tugs, workboats and multcats
- 10.6 Pusher tugs including combination units
- 10.7 Line handling craft



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